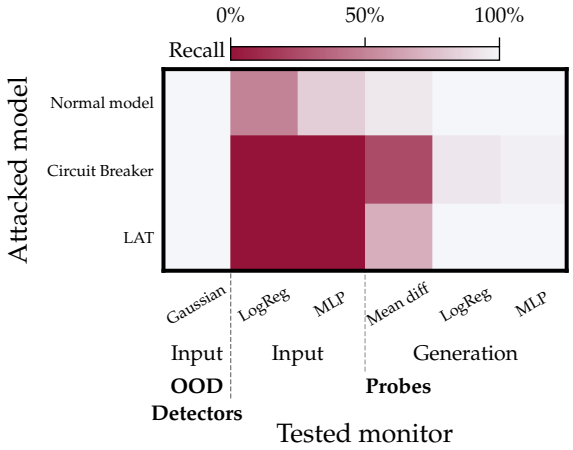


Figure 5: **Jailbreaking models trained using latent space methods bypasses monitors even without an explicit obfuscation loss.** This heat map shows the performance of monitors trained on different models trained using latent space methods against universal jailbreak embedding suffix attacks with no explicit obfuscation loss. Each row indicates what model (normal being Llama-3-8b-Instruct, full results in Fig. 4b) the suffix was trained on. Each column corresponds to a monitor we test this suffix against.

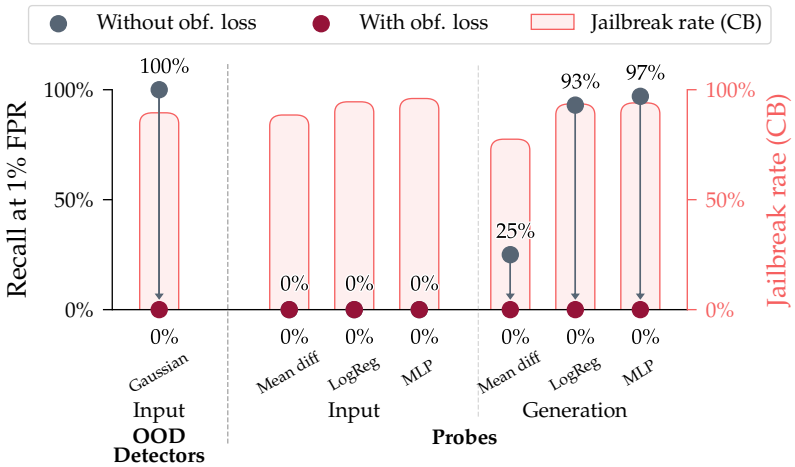


Figure 6: **Obfuscation attacks work against the circuit breakers model.** **Baseline probe recall** against attacks without explicit obfuscation loss are low, often 0% (as shown also in row 2 of Fig. 4b). For probes that perform well on the baseline, we can still obfuscate against them with **obfuscation attacks** while still achieving a high **jailbreaking rate** of over 75%. Note that OOD detector baselines perform better than with the normal model in Fig. 4b, likely because representation rerouting training makes harmful activations more unusual. We show the corresponding plot for the LAT model, with similar trends, in Fig. 19.

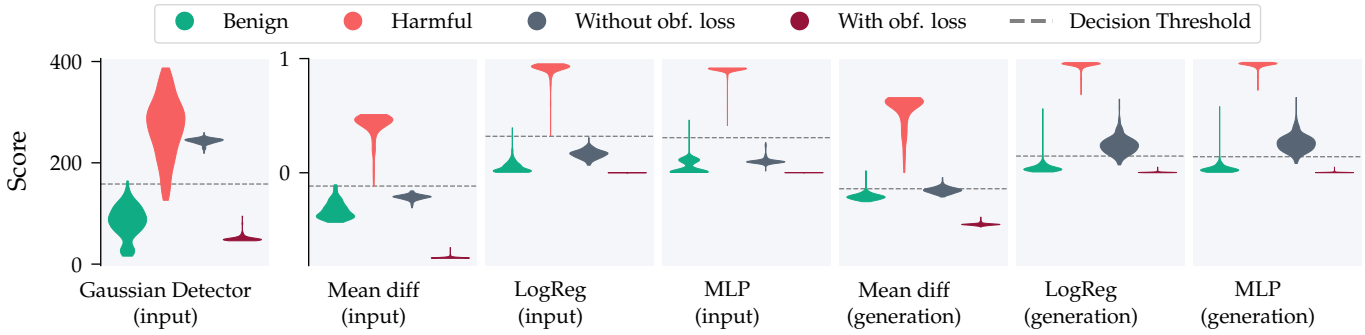


Figure 7: **The distribution of scores on inputs with standard jailbreak embedding suffixes is much closer to the benign distribution than the original harmful inputs.** We show the distribution of monitor scores on the circuit breakers model. While recall at 1% FPR does not show much obfuscation in Fig. 5, the raw harmfulness monitor scores show that jailbreaking circuit breakers also leads to significant obfuscation against generation-based probes.

the latent space. In this section we test if such training makes harmfulness representations less malleable, and thus if these models are harder to obfuscate against.

We begin by training obfuscation embedding space attacks (with and without an explicit obfuscation loss) against both models (each is based on Llama-3-8B-Instruct) in the same manner as §3.3. We present our results for the circuit breakers model in Fig. 6, and the LAT model in Fig. 19 of the Appendix. We find that we can induce obfuscated activations against both the LAT and circuit breakers models.

We now turn our attention to behavior only attacks (without an explicit obfuscation loss). Focusing on the circuit breakers model, we were able to achieve a high jailbreak success rate of 95% over the harmful test set using a universal embedding suffix (without an obfuscation loss). We attribute this success, as opposed to the low success rate of embedding attacks found by Zou et al. [2024], to optimizing the embedding to match a long harmful completion as opposed to a simple affirmative response.

To diagnose where the circuit breaker mechanism was failing, we use the same training procedure for monitors as §3.1, only applied to a circuit breaker model. Interestingly, we find that an embedding suffix trained to jailbreak this model *without any obfuscation loss* already obfuscates activations against many monitors (Fig. 5). This suggests that the reason our embedding suffix can jailbreak the model may be that it obfuscates its activations so that whatever scrambling mechanism circuit breakers introduced does not recognize the activations as harmful. Fooling this "implicit harmfulness detector" then transfers to fooling explicit external detectors, similar to our findings in Fig. 4c. We observe a similar phenomenon for the LAT model.

The normal embedding suffix (without explicit obfuscation) attack does not always obfuscate activations. In particular, the input-based OOD detector actually performs much better on the circuit breakers model than the normal one. The normal embedding suffix also does not obfuscate against generation-based probes, though it does significantly reduce the monitor harmfulness scores as Fig. 7 shows. We repeat, however, that an explicit obfuscation attack can fool all probes, including OOD detectors, as shown in Fig. 6.

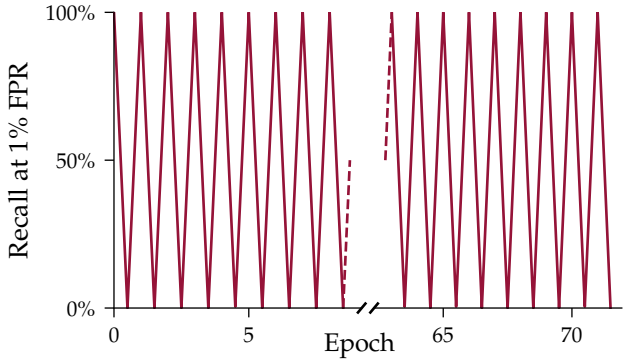
3.5 Adversarial Training Against Obfuscation Attacks

We have seen that all traditional latent-space monitors are highly vulnerable to obfuscated activations. It may be, however, that we can map out all the different ways a model can represent harmfulness with a large enough training dataset. We study question (3), *how many different ways can a model represent harmfulness*, by adversarially training monitors with obfuscated representations.

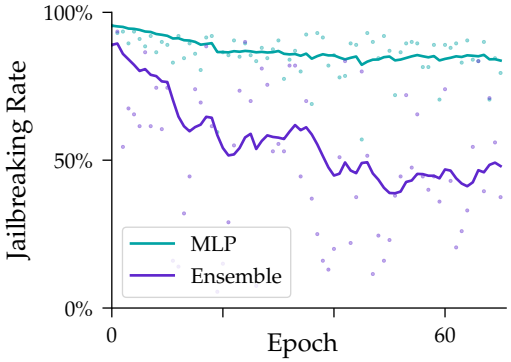
We let the attacker and defender take turns: in each epoch, the attacker trains an embedding attack to jailbreak the model and fool the current monitor, then the defender retrains a monitor against the obfuscated activations from the current and all previous attacks. We focus our attention to input based probes.

For the defender to benefit from adversarial training, the probes need to have enough capacity to guard against all previous attacks. We thus focus on the MLP probe, as well as a new probe that consists of an *ensemble* of mean difference probes. See Appendix A for details.

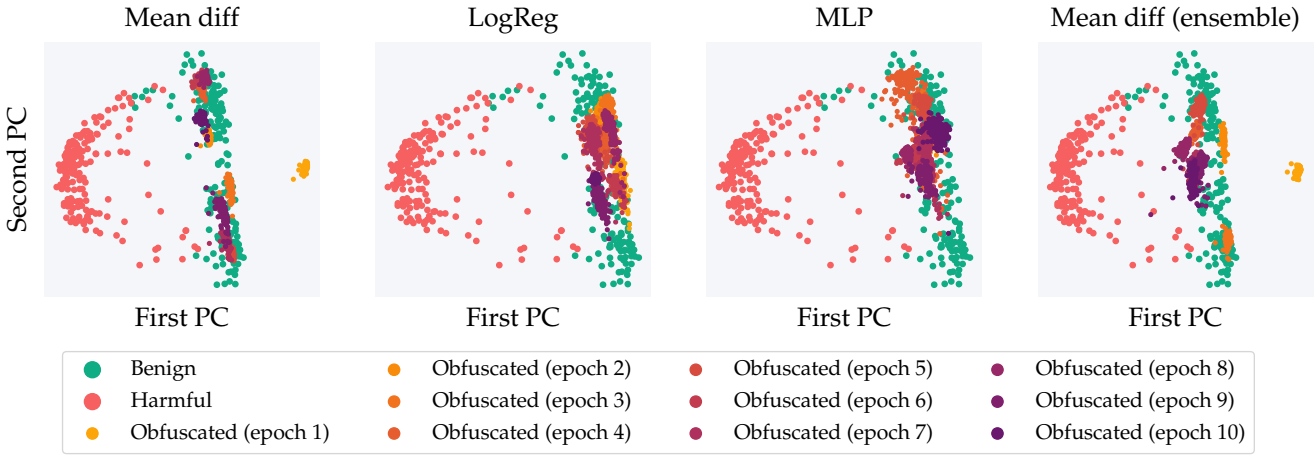
The attacker can consistently bypass adversarially trained monitors. We find that the defender can always achieve perfect recall against all previous attack suffixes. However, this does not mean the probe is robust to new attacks; indeed, even after over 70 epochs, the attacker can still find a



(a) **Even after 70 epochs, the attacker can still find new obfuscated activations that fool the monitor.** We show the monitor's recall (higher is better) as attacker and defender take turns (MLP probe; ensemble of linear probes shows the same trend in Fig. 20 of the Appendix). Recall is calculated on the most recent obfuscation attack (see Fig. 9 for performance on all previous obfuscation attacks). Corresponding jailbreaking rates for each attack are shown in Fig. 8b.



(b) **Throughout adversarial training, obfuscation attacks continue to jailbreak the model**. We show jailbreaking rate over time for the obfuscated attacks shown in Fig. 8a. Over time jailbreaking rate stays high for MLP, and become noisier for Ensemble but seem to plateau at an average of around 50%.



(c) **Over the course of adversarial training, obfuscated activations move around within the cluster of benign activations to successively evade each new defense.** We show the PCA visualization of activations from layer 22 and final token position of **benign**, **harmful**, and harmful + obfuscation embedding suffix inputs. PCs computed using benign and harmful activations.

Figure 8: **Results from adversarially training monitors.** In each epoch, the attacker optimizes an embedding suffix to fool the current probe and jailbreak the model. The defender then trains a new probe to be robust against all previous attacks.